

VANUA BALAVU, Fiji

WMO: 916760

Lat: 17.233S

Lon: 178.950W

Elev: 171

StdP: 14.61

Time Zone: 12.00 (PCF)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	70.4	71.3	56.9	69.5	73.3	58.8	74.4	73.7	25.9	73.2	24.9	73.5	12.7	130	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	87.8	80.3	86.8	79.8	85.9	79.4	83.6	85.1	82.3	84.5	81.2	83.8	9.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
83.3	175.0	84.4	81.7	166.1	83.2	80.6	159.8	82.6	48.0	85.0	46.3	84.7	45.2	83.7	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	21.4	19.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.9	81.2	81.7	81.6	80.4	78.4	77.1	76.0	76.3	76.7	77.8	79.3	80.5
	DBStd	2.84	1.97	1.89	1.67	1.91	1.89	1.72	1.95	2.29	2.24	2.14	1.72	2.13
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10548	967	887	981	913	880	812	807	816	800	862	879	944
	CDD65	5073	502	467	516	463	415	362	342	351	350	397	429	479
	CDH74	40664	5116	4943	5403	4394	3066	2031	1483	1650	1834	2595	3579	4570
	CDH80	6695	1199	1283	1330	781	254	67	41	90	85	227	426	912
(14) Wind	WSAvg	9.0	7.0	6.3	7.1	8.8	8.7	9.0	11.3	11.6	11.2	10.5	8.8	8.0
(15) Precipitation	PrecAvg	71.3	9.3	7.6	8.4	7.5	5.6	4.3	2.8	2.9	3.7	5.2	5.7	8.2
	PrecMax	95.4	19.1	19.0	19.6	16.8	10.4	12.6	5.0	6.9	9.2	11.7	13.1	15.9
	PrecMin	48.8	1.8	0.8	0.9	0.6	1.5	0.6	0.3	0.7	0.9	1.1	1.6	1.1
	PrecStd	12.6	4.6	4.4	4.3	3.9	2.7	2.9	1.3	1.7	2.2	2.5	3.0	3.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.7	89.1	88.8	87.6	85.3	83.4	82.7	84.6	83.7	86.0	86.1	88.4
		MCWB	80.7	81.6	81.2	80.7	79.1	77.1	75.1	78.2	76.6	79.3	78.6	80.5
	2%	DB	87.1	87.3	87.1	86.0	83.7	81.7	81.2	82.4	81.9	83.8	84.6	86.6
		MCWB	79.8	80.2	80.2	79.9	77.0	76.2	74.9	76.8	75.5	77.2	77.8	79.6
	5%	DB	86.0	86.4	86.1	84.9	82.5	80.6	80.1	80.8	80.9	82.2	83.4	85.3
		MCWB	79.4	79.6	79.6	78.9	75.8	75.1	74.5	75.3	74.8	76.0	77.2	78.9
	10%	DB	84.9	85.4	85.2	83.8	81.4	79.6	79.1	79.5	79.9	81.0	82.4	84.1
		MCWB	78.8	79.0	78.9	77.9	75.4	74.4	73.9	74.0	74.3	75.1	76.5	78.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	84.9	84.8	84.7	84.1	80.7	79.1	79.6	79.6	79.1	79.7	82.5	84.1
		MCDB	85.7	86.2	85.9	85.2	82.8	81.4	80.0	83.0	81.9	84.4	83.6	85.0
	2%	WB	82.9	82.8	82.5	82.1	78.8	77.7	77.8	77.9	77.5	78.3	80.7	81.9
		MCDB	84.6	85.2	85.0	83.8	82.0	80.2	78.5	81.2	80.2	82.1	82.3	84.1
	5%	WB	81.2	81.5	81.2	80.8	77.6	76.9	76.5	76.4	76.3	77.4	79.2	80.7
		MCDB	84.0	84.4	84.2	82.7	81.0	79.3	78.0	79.6	79.3	80.9	81.1	83.1
	10%	WB	79.8	80.2	80.0	79.7	76.7	76.1	75.3	75.3	75.4	76.6	78.2	79.5
		MCDB	83.5	83.6	83.5	82.3	80.2	78.7	78.1	78.4	78.5	79.9	80.7	82.5
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.0	6.9	6.2	5.6	4.9	5.1	5.2	5.4	5.8	5.9	6.3
		MCDBR	7.9	7.9	7.7	7.1	6.8	5.7	6.1	6.5	6.4	6.8	6.9	7.5
	5% WB	MCWBR	4.2	4.3	4.1	4.1	3.7	3.4	3.6	3.4	3.7	3.8	3.8	4.2
		MCWBR	7.2	7.4	7.5	7.0	6.3	5.7	5.4	6.3	5.9	6.4	6.3	6.9
(40) Clear-Sky Solar Irradiance	taub		0.385	0.380	0.377	0.388	0.386	0.387	0.379	0.385	0.400	0.388	0.385	0.388
		taud	2.630	2.655	2.666	2.624	2.608	2.589	2.598	2.580	2.538	2.576	2.607	2.606
	Ebn at Noon		305	303	297	282	271	265	271	279	285	297	302	303
		Edh at Noon	32	31	30	29	28	28	28	30	33	33	33	33
(44) All-Sky Solar Radiation	RadAvg		1999	1919	1795	1536	1359	1217	1309	1483	1691	1861	1932	1942
	RadStd		186	168	129	122	96	97	67	130	116	141	166	136

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page